

When to Use

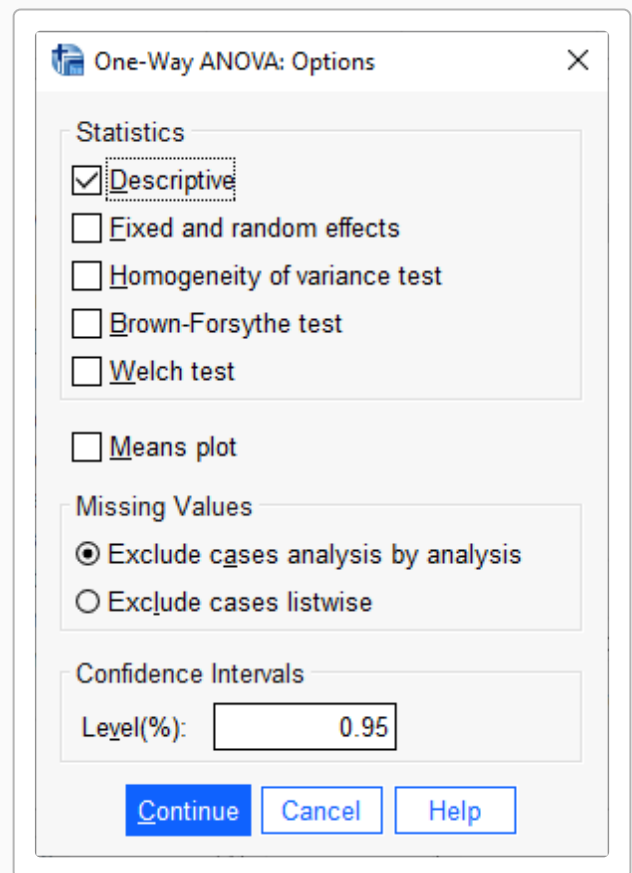
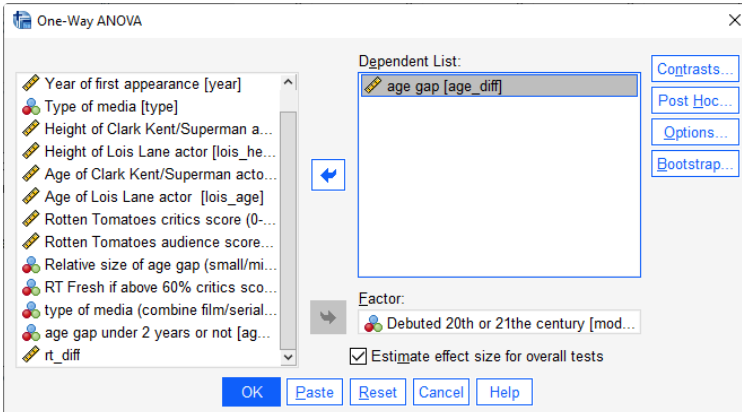
To compare means of a quantitative variable obtained from 2 or more independent groups.

H Research Hypothesis: There are significant differences in emotional impact scores among the three age difference categories (Minimal, Average, Big), with portrayals featuring larger age differences receiving higher emotional impact ratings.

H Null Hypothesis: The three different age difference categories have the same mean emotional impact score.

Analyze → General Linear Model → Univariate

- Highlight the “Dependent” variable (be sure it is **quantitative**) and click the arrow
- Highlight the “Factor” (IV, grouping) variable (be sure it is **qualitative**) and click the arrow
- “Options” — check that you want “Descriptive Statistics”
- “Post Hoc” — highlight the factor and click the arrow



Note

You can also perform this analysis using the “ONEWAY” procedure we used for the 2 BG ANOVA. It has the same post hoc analysis choices and produces equivalent output.

</> SPSS Syntax

```
UNIANOVA emotional_impact BY age_grp
  /METHOD=SSTYPE(3)
  /POSTHOC=age_grp(TUKEY LSD)
  /PRINT=DESCRIPTIVE
  /CRITERIA=ALPHA(.05).
```

①
②
③
④
⑤

- ① DV "by" IV
- ② Appropriate for =/n
- ③ IV & enter Post Hocs
- ④ Get descriptive stats
- ⑤ Sets p-value for the Post hoc tests

SPSS Output

Descriptives

emotional_impact

				95% Confidence Interval for Mean				
	N	Mean	Std. Deviation	Std. Error	Lower Bound	Upper Bound	Minimum	Maximum
Minimal	12	13.08	3.00	0.87	11.18	14.99	8.00	18.00
Average	16	14.62	2.92	0.73	13.07	16.18	10.00	19.00
Big	18	12.39	3.73	0.88	10.53	14.24	4.00	18.00
Total	46	13.35	3.35	0.49	12.35	14.34	4.00	19.00

ANOVA

emotional_impact

	Sum of Squares	df	Mean Square	F	Sig.
Between Groups	43.490	2	21.745	2.020	.145
Within Groups	462.944	43	10.766		
Total	506.435	45			

Multiple Comparisons

Multiple Comparisons

Dependent Variable: emotional_impact — LSD

			95% Confidence Interval			
(I) age_grp	(J) age_grp	Mean Difference (I- J)	Std. Error	Sig.	Lower Bound	Upper Bound
Minimal	Average	-1.542	1.253	.225	-4.069	0.985
Minimal	Big	0.694	1.223	.573	-1.772	3.161
Average	Minimal	1.542	1.253	.225	-0.985	4.069
Average	Big	2.236	1.127	.054	-0.037	4.510
Big	Minimal	-0.694	1.223	.573	-3.161	1.772
Big	Average	-2.236	1.127	.054	-4.510	0.037

Multiple Comparisons

Dependent Variable: emotional_impact — Tukey HSD

The p-value of 0.145 means that there is about a 14.5% chance that this result is a Type I error.

Remember, even if the printout shows $p = .000$, never report $p = .000$, because that would suggest there is no possibility of a Type 1 error. Instead, report " $p < .001$ ".

Notice that each pairwise comparison is presented twice!

Minimal vs Average = Average vs Minimal

Be sure you get the direction of each significant mean difference right!!

Minimal = Average

Minimal = Big

Average = Big

These LSD p-values can also be used for Bonferroni tests. Had we been interested in only the comparison of Minimal v Average & Average v Big, we would want to test each using $p = .05 / 2 = .025$.

Notice that each pairwise comparison is presented twice!

Minimal vs Average = Average vs Minimal

					95% Confidence Interval	
(I) age_grp	(J) age_grp	Mean Difference (I- J)	Std. Error	Sig.	Lower Bound	Upper Bound
Minimal	Average	1.542	1.253	.442	-1.500	4.583
Average	Minimal	-1.542	1.253	.442	-4.583	1.500
Minimal	Big	-0.694	1.223	.838	-3.663	2.274
Big	Minimal	0.694	1.223	.838	-2.274	3.663
Average	Big	-2.236	1.127	.129	-4.973	0.501
Big	Average	2.236	1.127	.129	-0.501	4.973

Be sure you get the direction of each significant mean difference right!!

Minimal = Average

Minimal = Big

Average = Big

Usually, but not always, results from LSD & HSD analyses agree. When they don't, the source of the disagreement is usually that the study is "underpowered" for the HSD. A sample size providing adequate power to reject H_0 using the more sensitive LSD may not provide adequate power to reject H_0 using the more conservative HSD. This most likely to happen when the sample size is selected based on $p < .05$, and then the HSD is applied.

Post Hoc "Computators"

SPSS does not provide post hoc analyses for all ANOVA models (e.g., WG designs). Also, there may be occasions when you want to compare means from a study that didn't post analyses, or did them differently than you would have preferred. One additional advantage of using these is that you can provide your readers with the LSD or HSD values that were the basis of your post hoc tests.

Steps for computing LSD minimum mean difference

1. Determine df(error) for the analysis: **df(error) = 43**
2. Use the t-table to determine the critical value of t (at $p = .05$) for this df(error): **t-critical = 2.017**
3. Determine the MSerror for the analysis: **MSerror = 10.766**
4. Determine the average number of data points upon which each mean is based: **n = N/k = 46/3 = 15**
5. Apply the LSD formula to obtain the minimum mean difference:

$$d_{LSD} = t \times \sqrt{\frac{2 \times MS_{error}}{n}} = 2.017 \times \sqrt{\frac{2 \times 10.766}{15}} = 2.3898$$

Steps for computing HSD minimum mean difference

1. Determine the number of conditions: **k = 3**
2. Determine df(error): **df(error) = 43**
3. Use the q-table (Studentized Range Statistic) for k and df(error): **q-critical = 3.433**
4. Determine the MSerror: **MSerror = 10.766**
5. Determine n: **n = 15**
6. Apply the HSD formula:

$$d_{HSD} = q \times \sqrt{\frac{MS_{error}}{n}} = 3.433 \times \sqrt{\frac{10.766}{15}} = 2.8766$$

6. Compare the minimum mean difference with each pairwise mean difference:
 - If the pairwise mean difference is **larger** than the minimum mean difference, then those two conditions have means that are "significantly different"
 - If the pairwise mean difference is **smaller** than the minimum mean difference, then those two conditions have means that are "statistically equivalent"

Applying LSD/HSD values to the pairwise comparisons

Applying these LSD/HSD values to the pairwise comparisons... Minimal = 13.08 Average = 14.62 Big = 12.39

	Minimal v Average	Minimal v Big	Average v Big
Mean Difference	1.54	0.69	2.24
LSD Result	=	=	=

	Minimal v Average	Minimal v Big	Average v Big
HSD Result	=	=	=
RH:	<	<	<
RH: support?	Not supported	Not supported	Not supported

Reporting the Results

 Sample APA Write-Up

The emotional impact scores for each age difference category are summarized in the table above. Minimal ($M = 13.08$, $SD = 3.00$), Average ($M = 14.62$, $SD = 2.92$), Big ($M = 12.39$, $SD = 3.73$). There were no significant mean differences in emotional impact among the three age difference categories, $F(2,43) = 2.02$, $MSe = 10.77$, $p = 0.145$. Pairwise comparisons using LSD (with a minimum mean difference = 2.39) revealed that there was no difference between the average emotional impact of Minimal and Average portrayals or between the average emotional impact of Minimal and Big portrayals or between the average emotional impact of Average and Big portrayals.

Key Reporting Elements

When reporting k BG ANOVA results with post hocs, be sure to communicate:

- The research hypothesis (if there is one)
- Descriptive statistics (M, SD) for each group
- The overall ANOVA results (F , df , p , MSe)
- The post hoc method used and the minimum mean difference value
- The direction of each significant pairwise difference
- Which pairwise comparisons were not significant
- Whether or not results support the research hypothesis